Cardiovascular Updates

Cardiovascular Therapeutics

For all cardiovascular therapeutics chapters (e.g., hypertension, heart failure, angina, acute coronary syndrome, etc.) ... please see the new chapters and the new videos on the portal

DRUG INFORMATION

Page 57 – Digoxin monograph

Female gender is a risk factor for digoxin toxicity

Page 76 – Statins

Factors that ↑ the risk of myotoxicity: add the following new factors

- Patient's factors: ADD – including history of unexplained muscle/joint/tendon pain), and nocebo effect.
- Disease states: ADD – HTN, HF, and neuromuscular diseases
- Drug interactions: ADD – warfarin, cocaine, and amphetamines

Infectious Diseases Updates

Vaccines

Page 86 table – yellow fever vaccine (YF-Vax®) is an example for live-attenuated vaccine

Page 89 – some vaccination rules

Oral and intranasal vaccines: they can be given at the same time as, or any time before or after any other live-attenuated or inactive vaccine, regardless of the route of administration of the other vaccine. These are general rules, but if there are specific restrictions, we must follow them. For example, live attenuated oral typhoid vaccines should be separated by at least 8 hours from live attenuated cholera vaccine

Community Acquired Pneumonia

Please, see the new chapter and videos on the portal

Acute Otitis Media and Otitis Externa

Page 97 – Notes

- In case of treatment failure noted on days 10-28, use an antibiotic that is not used in the same infection
- Macrolides are reserved for patients with ***anaphylactic*** reactions to beta lactams and must ensure anaphylaxis before offering macrolides for acute otitis media

Page 98 – Chart

- If no improvement by day **3-9** or recurrence by day 10-28 (for the whole chart)
- recurrent acute otitis media is defined as (≥ 3 episodes in 6 months or ≥ 4 episodes in 12 months)

Tuberculosis

Please, see the new chapter ad videos on the portal

Bacterial Skin Infections

Page 114 – Impetigo

- **Clinical presentation:** superficial skin infection primarily affecting children. It can be Primary (affecting intact skin) or secondary (affecting non-intact skin). Impetigo can be classified into:
  - **Nonbullous:** more common – caused primarily by S. aureus and/or S. pyogenes – papules, vesicles, and pustules, and dry to honey-coloured crust – spontaneously resolve in 2-3 weeks with no scarring – affecting face and extremities. Complications are rare (e.g., glomerulonephritis and cellulitis)
  - **Bullous:** caused by toxin-producing S. aureus – fluid-filled vesicles that rupture easily– brown crust – spontaneously resolve in 2-3 weeks with no scarring – affecting the trunk and body folds. Complications are infrequent (e.g., glomerulonephritis and cellulitis)
  - Ecthyma: deeper nonbullous impetigo – skin ulceration
- **Causative:** S. aureus is the most common (S. pyogenes is more common in ecthyma)
- **Transmission:** to close contacts and other body areas – noncontagious once crust or 48 hours after staring antibiotic therapy
- **Risk factors:** neonates and young children, broken skin, close contact with patients with active lesions, hot/humid weather, comorbidities (HIV, diabetes, malnutrition), and carries of S. aureus
- **Management**
  - **Non-drug measures:** apply normal saline or warm water compress for 10-15 minutes 3-4 x/day to remove crust and promote healing
  - Manage any skin barrier defects such as abrasion, eczema, insect bites, etc.
  - **Nonbullous and localized/mild bullous: topical** mupirocin of fusidic acid 2-3x/day or ozenoxacin 2x/day x 5 days → if NO improvement after 48 hours or develop systemic infection (e.g., fever), **oral antibiotic** (cloxacillin, cephalexin, doxycycline, clindamycin, or erythromycin) x 7 days and swab exudate for C&S → if NO improvement after 48 hours, modify therapy based on C&S results
    - **MRSA suspected:** doxycycline, clindamycin, and SMX/TMP
    - **Candidates for initial oral antibiotics:** widespread infection, immunocompromised patients, and severe infections (extensive crusting, ulceration under the crust, fever, or bacteremia)
  - Recurrent impetigo (possible nasal carrier): nasal swab for S. aureus. If decolonization is required, apply mupirocin OR fusidic acid into nares 3x/day x 3 days. Wash nares, axilla, and perineum with chlorhexidine

Page 114 – Folliculitis

- **Clinical presentation:** itchy, painful, and inflamed hair follicle that is surrounded by erythema. It can be infected or non-infected (pseudo folliculitis)
- **Causative:** S. aureus (more common) or P. aeruginosa (hot tub rash)
- **Risk factors:** friction, heat, and moisture
- **Non-drug measures:** avoid friction, and skin occlusion
- **Management**
  - **Localized:** avoid friction and occlusion x 7 days → use hot compress/ drying agent/ antiseptic x 7 days → **topical** mupirocin, fusidic acid, or clindamycin solution x 7 days → **oral antibiotic** and swab the exudate for C&S test
    - **First choice:** cephalexin x 7 days
    - **Second line:** SMP/TMP, doxycycline, clindamycin, or cloxacillin x 7 days

- **MRSA suspected:** SMP/TMP or doxycycline x 7 days
- **Hot tub folliculitis:** ciprofloxacin
- If no improvement in 2 days, modify therapy based on C&S results
- If resistant/recurrent → treat for MRSA

- Start with oral antibiotic if extensive, systemic symptoms, or immunocompromised

Page 114 – Erysipelas

- **Management**
  - Mild and no systemic symptoms
    - **First line:** po penicillin VK or amoxicillin x 5 days
    - **Second line:** po cefuroxime x 5 days (for penicillin allergy)
    - **Third line:** po clindamycin x 5 days (used in cefuroxime allergy)
  - Moderate to severe symptoms (blistering, severe pain, progressing rapidly) or systemic symptoms (e.g., fever, heart rate > 100 bpm, leukocytosis, respiratory rate > 24 bpm, or hypotension) → rule out necrotizing infection (other symptoms: numbness, grey appearance) before starting antibiotics
    - **First line:** IV ampicillin or penicillin G
    - **Second line:** IV cefazolin (for penicillin allergy)
    - **Third line:** IV ceftriaxone (for cefazolin allergy)
    - **Allergy to cefazolin and ceftriaxone:** IV vancomycin (also used if no response to beta lactams) or clindamycin
    - The duration of treatment for above regimens is 5 -10 days, but the duration depends on response, complications, and risk factors

Consider compression stocking if leg is involved→ reduces recurrence

Page 114 - Cellulitis

Non-purulent	**Mild** ▪ **First line:** oral cephalexin x 5 days (in adults and children) ▪ **Cephalexin allergy:** oral clindamycin x 5 days (in children) and oral cefuroxime x 5 days (in adults) ▪ **Cephalexin and cefuroxime allergy <u>in adults</u>:** oral clindamycin x 5 days or cloxacillin x 5 days ▪ If no improvement in 48 hours, treat as moderate to severe **Moderate to severe** ▪ **First line:** IV cefazolin +/- probenecid x 5-10 days (in adults and children) ▪ **Second line or MRSA is suspected:** IV vancomycin x 5-10 days or IV/PO clindamycin x 5-10 days (in adults and children), or ceftriaxone x 5-10 days (in adults) ▪ **Third line:** IV cloxacillin x 5-10 days <u>in adults only</u> ▪ If no improvement in 48, assess for abscess → consider more aggressive treatment

<table>
<tr>
<td>Purulent (all ages) – incision and drainage C&S + empiric treatment</td>
<td>Mild
▪ MSSA:
o First line: po cephalexin x 5 days
o Cephalexin allergy: po cefuroxime x 5 days
o Cephalexin/cefuroxime allergy: clindamycin x 5 days
▪ MRSA: add po SMX/TMP or po doxycycline to MSSA regimen
▪ If no improvement in 2-3 days or slow progression: modify therapy according to the C&S results or treat as moderate-severe

Moderate – severe: systemic symptoms (e.g., fever, HR > 90, RR > 24, leukocytosis, hypotension, organ dysfunction), immunocompromised, deeper infection (e.g., skin sloughing)
▪ MSSA: IV cefazolin, IV clindamycin, or IV cloxacillin (all x 5-10 days)
▪ MRSA or penicillin allergy: IV cefazolin + po SMX/TMP x 5-10 days
▪ IV vancomycin (linezolid is alternative) x 5-10 days is reserved for severe cases
If no improvement in 2-3 days: modify therapy according to the C&S results</td>
</tr>
<tr>
<td colspan="2">▪ Patients with multiple recurrences may receive prophylactic antibiotic (controversial)
▪ Symptoms may appear to get worse in the first 24-48 hours before improving</td>
</tr>
</table>

Page 115 – Furuncle

- **Management**
  - **Non-drug measures:** warm water or saline compress 20-30 minutes 3-4 times daily, and apply transmission prevention strategies (see general non-drug measures)
  - Surgical drainage in **all** patients and cover the area after drainage.
  - Inadequate response to drainage, systemic symptoms (e.g., fever, tachycardia, tachypnea, leukocytosis, or hypotension), immunocompromised, complications (e.g., bacteremia), multiple/extensive/difficult to drain lesions**:** C&S testing and start empiric antibiotic therapy.
    - **If MSSA is suspected:** oral cephalexin, cloxacillin, clindamycin, doxycycline, or SMX/TMP x 5-7 days
    - **If MRSA is suspected:** oral SMX/TMP or doxycycline x 5-7 days
    - If no response to treatment, modify therapy according to C&S results
    - Patients who fail oral therapy may require IV therapy

If recurrent infection or spreading to household: use topical mupirocin intranasally 5 days per month x 3 months + chlorhexidine wash for 5-14 days

Page 116 – Dogs, cats, and human bites

Prophylaxis	Treatment
▪ **Candidates:** Cats bites, moderate (systemic symptoms exist) or severe (limb threatening) bite, immunocompromised patients, bites of face, feet, hands, or joints, or genitalia, deep bites involving bones, puncture wound, and edematous wound ▪ Start prophylaxis within 12 – 24 hours the of bite. **Prophylactic Regimen** ▪ **First line:** oral amoxicillin/clavulanic x 3-5 days ▪ **Amoxicillin allergy:** oral cefuroxime + metronidazole x 3-5 days ▪ **Third line:** oral doxycycline + metronidazole x 3-5 days	**Superficial infection** ▪ **First line:** oral amoxicillin/clavulanic x 7-10 days ▪ **Amoxicillin allergy:** oral cefuroxime + metronidazole x 7-10 days ▪ **Third line:** oral doxycycline + metronidazole x 7-10 days **Moderate infection** ▪ IV ceftriaxone + oral metronidazole **Severe infection** ▪ **Firs line:** IV piperacillin/tazobactam x 7-14 days ▪ **Penicillin allergy:** IV meropenem x 7-14 days ▪ Stepdown from IV to PO after 2-4 days (if CBC and CRP improve)

Travelers' Diarrhea

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Severity	Symptoms	Treatment
Mild	Diarrhea is tolerable, not distressing, and does not interfere with planned activities	Resolve within 24 hours Safe fluids +/- loperamide and/or Bi-subsalicylate If symptoms last > 2 weeks → further investigations
Moderate	Distressing diarrhea that interferes with planned daily activities	ORS +/- antibiotic +/- loperamide If symptoms last > 2 weeks → further investigations
Severe	Incapacitating diarrhea that causes the patient to cancel daily activities	ORS + antibiotic. May use loperamide in severe, but not dysentery If symptoms last > 2 days while on antibiotic → further investigations
Dysentery	Passage of bloody stool	

Infectious Diarrhea

Page 153

Management of Antibiotics-induced Diarrhea

The duration of therapy in the table should be 10 days

Page 154

Management of Recurrence of Antibiotics-induced Diarrhea

- Recurrence is defined as C. Difficile-induced diarrhea within 8 weeks of completing therapy
- **First recurrence (second episode):**

Treatment of first episode	Treatment of first recurrence
Metronidazole	Vancomycin 125 mg po QID x 10 days
Vancomycin	Fidaxomicin 200mg po BID x 10 days
Vancomycin standard regimen	Tapered and pulsed vancomycin regimen: 125mg po QID x 10-14 days, then BID x 7 days, then QD x 7 days, then every 2 – 3 days for 2 – 8 weeks

- **Second recurrence (third episode):**
    - Tapered and pulsed vancomycin regimen (see above)
    - Vancomycin 125 mg po QID x 10 days → fidaxomicin 400mg po TID x 20 days
    - OR fidaxomicin 200mg po BID x 10 days
- **Third recurrence (forth episode):** As under the second recurrence + consider microbiota transplantation

Endocarditis

Page 166 – table for surgical procedures requiring prophylaxis

Dental procedures warrant prophylactic therapy	Patients at risk of negative endocarditis outcome
▪ Dental work involves penetration of gum, periapical region of teeth, and oral mucosa. Examples include tooth extraction, drainage of dental abscess, dental cleaning, scaling, root planning, root canal treatment beyond the apex, reimplantation of avulsed teeth, intraosseous or intra-ligamentary injections, and periodontal procedures if bleeding anticipated **This does NOT include** dental anesthetic injections through non- infected tissue, routine filling, suture removal, taking dental radiograph, oral impression, restorative dental work such as placement, adjustment, or removal of prosthodontic or orthodontic appliances (e.g., crown, denture, braces), simple root canal treatment, rubber band placement, fluoride treatment, shedding of deciduous tooth, and bleeding due to minor lip trauma or trauma to oral mucosa	▪ Patients with prosthetic cardiac valves or valve repair using prosthetic materials +/- rheumatic heart disease or left ventricular assist device ▪ Previous, recurrent, or relapse of infectious endocarditis ▪ Cardiac transplant recipients + who develop valve regurgitation ▪ Congenital Heart Disease (CHD): o Unrepaired cyanotic CHD or o Within 6 months of repairing a CHD using prosthetic material or device o Repaired CHD with residual defects at/near a prosthetic material or device **This does NOT include** mitral valve prolapse, previous CABG, patients with cardiac pacemakers or pacemaker insertion, congestive heart failure, calcified aortic stenosis, hypertrophic cardiomyopathy, patent ductus arteriosus, and surgical correction of aorta

Page 166 – Prophylactic Regimen

Standard: Amoxicillin PO 2g 1-hour before procedure

Allergy to penicillin

- **Non anaphylactic:** cephalexin (2 g – or equivalent dose of 1st or 2nd generation cephalosporin), azithromycin (500mg), clarithromycin (500mg), or doxycycline (100mg)
- **Anaphylactic:** same as above except NO cephalexin

Cannot Swallow

IV or IM ampicillin (2g), ceftriaxone (1g) or cefazolin (1g)

Allergy + cannot swallow

IV or IM cefazolin (1g), or ceftriaxone (1g)

NOTES

- Clindamycin is no longer recommended for prophylaxis
- If the patient is taking an antibiotic for endocarditis prophylaxis for another indication, select an agent from a different class OR delay the procedure for ≥ 10 days after finishing the antibiotic before giving the same antibiotic for prevention
- If the pre-surgery dose is missed, it can be taken up to 2 hours post-surgery
- Avoid intramuscular injections in patients on anticoagulant

Lyme Disease

Page 175 – table for management of Lyme disease

Treatment of Lyme Disease

Stage	Management
Single EM or multiple EM with no CNS, musculoskeletal, or CV symptoms	▪ **First choice:** doxycycline (all ages), amoxicillin, or cefuroxime – all x 14 days ▪ Azithromycin x 7 days (5-10 days) – only if other agents cannot be used. Test for cure
Facial nerve palsy	▪ **First choice:** doxycycline x 14-21 days for all age groups ▪ **Second line:** amoxicillin or cefuroxime – all for 14-21 days ▪ If diagnosis is uncertain, add steroid along with the antibiotic and D/C steroid once Lyme disease is confirmed
Meningitis or polyneuropathy	▪ **First choice:** IV ceftriaxone x 14 days (10-21 days) – is used empirically until other causes are excluded. It can be used for full 21 days. If the patient is mildly ill + significant response to antibiotic, may step down to doxycycline (total duration = 21 days) ▪ **Alternative:** doxycycline x 14-21 days ▪ **Second line:** IV cefotaxime or IV penicillin G x 14-21 days – may start empirically and then step down to PO doxycycline – see ceftriaxone
Carditis	▪ **First choice in mildly ill patients:** doxycycline, amoxicillin, or cefuroxime – all x 14-21 days (outpatient) ▪ IV ceftriaxone x 14-21 days is the **first choice** in patients with arrhythmia, cardiac involvement, and PR interval ≥ 300 millisecond (inpatients). Once stable, may step down to PO doxycycline for a total duration of 21 days
Arthritis	▪ **First choice:** doxycycline, amoxicillin, or cefuroxime – all x 28 days. Some residual swelling may remain after treatment and will take longer to resolve. ▪ **Alternatives:** IV ceftriaxone or IV cefotaxime – all x 14-28 days – consider if oral therapy fails. ▪ If synovitis persists x ≥ 2 months following doxycycline (treatment failure), or if symptoms reoccur, consider IV ceftriaxone for up to 28 days ▪ **Second line:** IV Penicillin G x 14-28 days 2nd line
Late neuroborre-liosis	▪ **First line:** IV ceftriaxone or cefotaxime x 14-28 days ▪ **Alternative:** doxycycline x 28 days ▪ **Second line:** IV penicillin G x 14-28 days
Pregnancy and breastfeeding	▪ No transmission and no risk to the embryo ▪ **Prevention:** DEET can be used ▪ **Post-exposure prophylaxis:** doxycycline x 1 dose ▪ **Mild or localized disease:** amoxicillin or cefuroxime (avoid doxycycline ▪ **Severe cases:** IV ceftriaxone of cefotaxime

- Doxycycline dose is 200mg for adults and patients > 45Kg (4 mg/kg/day for patients ≤ 45kg divided BID to maximum of 100mg – give once daily in neuroborreliosis)
- Within 24-hour of starting therapy, patients may experience benign Jarisch-Herxheimer-like reaction (fever, sweating, myalgia, arthralgia, chills, and rigors) due to the release of toxin associated with lysis of spirochetes.
- Antibiotic therapy is very effective, regardless of the stage, and treatment failure is rare. Symptoms completely resolve except nerve palsy

Page 176 – Post-Exposure Prophylaxis

- Estimated time of attachment ≥ 24 hours or if the tick is engorged (engorged tick = large tick of up to 1cm because of feeding – engorgement is a better predictor)

HIV

Page 181 – Drugs for Pre-Exposure Prophylaxis

- Tenofovir disoproxil fumarate (tenofovir DF) + emtricitabine (Truvada) once daily – not for patients with CrCl < 60 ml/min
- Tenofovir alafenamide (tenofovir AF) + emtricitabine (Descovy) once daily – not for patients with CrCl < 30 ml/min and not for PrEP in receptive of vaginal sex
- In patients with impaired kidney function who need individual agents for dose adjustment, use lamivudine instead of emtricitabine in combination with tenofovir
- Prophylaxis therapy should start before the exposure begins, and continue during exposure along with safe sex practices

Before starting treatment

- The patient must be HIV-negative. HIV-negative status must be maintained during treatment
- Th pre-exposure prophylaxis should start ASAP after the negative HIV test. If the treatment is delayed by > 2 weeks, repeat the test to make sure it is still negative
- If the HIV test is negative, but symptoms suggest HIV, defer the prophylactic therapy, and repeat the test after 1-3 weeks

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Regimens for Post-Exposure Prophylaxis

1st Choice	▪ Tenofovir DF/emtricitabine combination once daily (Truvada) + **EITHER** dolutegravir (once daily), or raltegravir (one or twice based on formulation). o **Other add on options for non-occupational exposure:** Darunavir/R ▪ If CrCl < 60: use zidovudine instead of tenofovir
Alternative	▪ Lamivudine once daily + tenofovir once daily + **EITHER** atazanavir/R once daily, or lopinavir/R once daily o **Other add on options for occupational exposure:** darunavir/R once daily, etravirine twice daily or rilpivirine once daily o **Other add on options for non-occupational exposure:** darunavir/C once daily, or raltegravir high dose ▪ Lamivudine/zidovudine combination twice daily (Combivir) + one of the above options ▪ Tenofovir DF/emtricitabine/elvitegravir/cobicistat combination once daily (Stribild) – not for CrCl < 70 mL/min

Cross Sensitivity of Beta Lactams

Reference: https://www.bugsanddrugs.org/8dbf3d51-1be5-4669-be8e-679fd1564cd6

Some examples:

- Safe Cephalosporins in patients with ALL beta lactams allergy: Cefazolin, cefixime, and ceftazidime
- Allergy to amoxicillin/ ampicillin: use all cephalosporins EXCEPT: cephalexin, cefprozil, cefaclor, and cefadroxil.
- Allergy to cloxacillin and piperacillin: safe to use all cephalosporins.

CNS Updates

DEPRESSION

See new chapter on the portal

BIPOLAR

NEW CONTENTS

Types of Bipolar

- Bipolar I: alternating mania + depression
- Bipolar II: hypomania + depression
- Rapid cycling: ≥ 4 cycles of manic/hypomanic + depressive episode over 1 year
- Substance-induced: symptoms appear soon after substance intoxication/withdrawal
- Switching: rapid change from depression to mania when starting therapy

Page 8

Non-Drug Measures

Address cognitive impairment: ensure full remission, D/C contributing drugs, treat concomitant disorders, and general self-care measures (e.g., healthy diet, exercise, adequate sleep)

Management During pregnancy

- Folic acid 0.4 – 5mg/day preconception x 3 months until end of 1st trimester… then 0.4mg until delivery and x 4 – 6 weeks postpartum
- Avoid therapy in 1st trimester if possible
- Preferred agents are lamotrigine and quetiapine. Alternative agents include, Li, CBZ, aripiprazole, and olanzapine.
- If on Li → monitor echocardiogram of embryo

Page 9

See updated management chart

NEW CONTENTS

Rapid Cycling

- Poor response
- Avoid antidepressants
- May require 3 mood stabilizers (e.g., Li + divalproex + SGA)
- **Refractory cases:** ECT, clozapine, or levothyroxine

Drug Selection Considerations

Balance adverse reactions & patient's characteristics

- Lithium: 1st choice of first line agents – ↓ suicidal ideation + minimal dementia. Caution with renal diseases
- Divalproex: teratogenic + menstrual irregularity
- CBZ: skin – induced adverse reaction + high potential for drug interactions
- Antipsychotics: high risk of metabolic syndrome

Acute Depressive Episode

D/C antidepressants if any (will worsen manic symptoms or cause rapid cycling)

- Assess risk to others (may hospitalize)
- Identify/treat complicating comorbidities
- D/C contributing drugs/substances

Acute Manic Episode

In addition to mood stabilizers, may add
- Clonazepam or lorazepam PO/IM for acute agitation
- Haloperidol PO/IM for acute psychosis

Patient is UNTREATED

- Start 1st line agent monotherapy. Restart previous successful therapy if any
- If severe symptoms: combine 1st line agents

Mania	**Monotherapy:** Li (most important), divalproex, quetiapine, risperidone, asenapine, aripiprazole, paliperidone) **Combinations:** Li OR divalproex + either quetiapine, risperidone, asenapine, or aripiprazole (start antipsychotic 1 – 3 days first)
Depression	**Monotherapy:** Li, lamotrigine, lurasidone, or quetiapine, cariprazine **Combination:** Li or divalproex + lurasidone

Patient is TREATED with 1st line agent

- Assess adherence (including blood level if on Li, divalproex, or CBZ – therapeutic level unestablished)
- **If NO adherence issues:** add or switch to another 1st line
- **Non-adherent:** adjust the dose

Try x 2 weeks (2 – 4 weeks for depression) then assess

- If symptoms persist while on monotherapy → Change to OR add a 1st line agent
- If symptoms persist while on a combination → Change one or both drugs to 1st line agents

Try x 2 weeks (2 – 4 weeks for depression) then assess

If symptoms continue → ADD/SWITCH to 2nd line agents **after trying multiple 1st line agents**

Mania 2nd line agents	**Monotherapy:** CBZ, olanzapine, ziprasidone, or haloperidol **Combinations:** Li + divalproex OR Lithium or divalproex + olanzapine **ECT:** for severe behavioral disturbances, psychosis, suicidal ideation, poor response to combination therapy, or not tolerated adverse reactions
Depression 2nd line agents	**Monotherapy:** divalproex Combination: Olanzapine + fluoxetine, and adjunctive SSRI or bupropion ECT: as under mania

Try x 2 weeks (2 – 4 weeks for depression) then assess

- Seek specialized care

Mania 3rd line agents	Adjunctive therapy with clozapine, CBZ, tamoxifen, calcitonin, levetiracetam, phenytoin, and omega-3
Depression 3rd line agents	Adjunctive therapy with SNRI, MAOI, IV ketamine, modafinil, typical antipsychotics, and light therapy. RxFiles: levothyroxine, CBZ (watch for interactions), and pramipexole

Maintenance Therapy

- Once a regimen is effective: Continue, may use lower dose, and prevent relapse. Combination is required if patient has multiple relapses
- Monitor effect in 2 months after starting or adjusting therapy
- If remains stable x 6 months: may ↓ the dose or D/C one agent

1st line maintenance	**Monotherapy:** Li, divalproex, quetiapine, asenapine, aripiprazole, or lamotrigine **Combination:** Li OR divalproex + Either quetiapine, or aripiprazole
2nd line maintenance	**Monotherapy:** olanzapine, risperidone, paliperidone, or CBZ **Combination:** Li or divalproex + lurasidone

Duration of therapy

- **Acute phase:** 2 weeks for mania and 2 – 4 weeks for depression
- **Maintenance:**
  - 1st episode: 1 year (min. 6 months)
  - 2nd episode: indefinite
  - Severe first episode or family history → indefinite

Avoid in Acute Mania: Monotherapy (allopurinol, gabapentin, lamotrigine, omega 3, and topiramate) or Combination (CBZ + risperidone or CBZ + olanzapine)

Avoid in Acute Depression: antidepressants monotherapy, gabapentin, aripiprazole or ziprasidone, mifepristone, and lamotrigine + folic acid

PSYCHOSES

No Updates

DEMENTIA

No Updates

SUBSTANCE ABUSE

No Updates

ANXIETY

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Obsessive Compulsive Disorders

- **Psychotherapy:** specialized CBT is important
- An SSRI in antidepressant doses is **1st choice – start low and ↑ slow - try full dose x 6 weeks.**
    - **If improved/much improved:** continue for full 12 weeks, then reassess.
        - **If ineffective:** Assess adherence, ↑ dose of SSRI to the maximum dose → if ineffective, switch to another SSRI, venlafaxine, or clomipramine
        - **If not tolerated:** try another SSRI, or switch to venlafaxine, or clomipramine
        - **If partially effective or fail 2 antidepressants:** augment with atypical antipsychotic (aripiprazole or risperidone)
    - **If ineffective:** augment with lamotrigine, memantine, olanzapine, or quetiapine
    - **If ineffective:** switch to mirtazapine monotherapy or augment with topiramate or amantadine
    - **If ineffective:** Triple drug therapy = SSRI or SNRI + atypical antipsychotic + a glutamate modulator (lamotrigine, memantine, topiramate, or amantadine)
- Continue effective regimen x 1 year and reassess (longer therapy may be needed. Relapse is common after discontinuation)
- Monitor the response to treatment after 4 weeks, and 12 weeks, and whenever making therapy decisions, using assessment scales such as Y-BOCS and 6-item brief OCD scale.
- BDZ are not effective as monotherapy unless there are comorbidities
- **Pregnancy:** Screen for OCD and psychosis pre-conception and postpartum. During pregnancy, Specialized CBT is the 1st choice. If symptoms are severe and associated with significant impairment → consider SSRI.
- **Breastfeeding:** specialized CBT is 1st choice. Escitalopram or sertraline can be used if severe

Post-Traumatic Stress Disorder

- Week 1 – 4: reassurance that symptoms are short term. Consider treatment if patient has significant impairment
- **Trauma-focused psychotherapy:** 1st line. Include Prolonged Exposure (PE), cognitive processing therapy (CPT), and Eye Movement Desensitization and Reprocessing Therapy (EMDR) are more effective than CBT. Psychotherapy may worsen comorbidities such as suicidal ideation, depression, and substance abuse, thus stabilization of comorbidities, using pharmacotherapy, is required before starting psychotherapy
- **Pharmacotherapy:** also 1st line especially when psychotherapy is not available or when stabilization is required. It is more effective than psychotherapy in military/combat-related trauma.
- **1st choice:** SSRI (fluoxetine, paroxetine, and sertraline) and venlafaxine.
- **2nd choice:** mirtazapine (helpful with insomnia and loss of appetite), moclobemide, phenelzine, and fluvoxamine
- **Antipsychotics:** SGA (especially quetiapine, aripiprazole, and olanzapine) are used for augmentation
- **Management approach:** use an antidepressant of 1st choice x 4 – 6 weeks. If ineffective → change to an agent from the same or different class x 4 – 6 weeks → If ineffective, switch to a different class or augment with SGA
- **BDZ** monotherapy and adjunctive therapy are not recommended (short term for acute anxiety/insomnia is OK in patients receiving psychotherapy)
- **Trazodone** improves sleep. **Prazosin** improves sleep and reduce nightmares, but its efficacy is questionable
- **Cannabis** has limited data and it is not recommended in PTSD
- **Prevention: Propranolol** can prevent PTSD if taken few hours before traumatic event and it helps bad memories to be more distant. A shot of **hydrocortisone** within 6 hours post traumatic event may prevent PTSD
- A combination of psychotherapy + pharmacotherapy is not more effective than either therapy alone. Consider patient's preference
- **Pregnancy:** Psychotherapy is 1st choice. If symptoms are severe or cause significant impairment, citalopram, escitalopram, and sertraline are used as 1st line but monitor neonates for withdrawal symptoms if used in 3rd trimester
- **Breastfeeding:** same as pregnancy

INSOMNIA

NEW AND REVISED CONTENTS

Clinical Presentation

- **Insomnia** can be short-term insomnia/acute (< 3 months) or chronic (≥ 3 months)
- It is commonly a symptom of other conditions rather than being a disease by itself

Assessment

- Sleep diaries for 1-2 weeks, Insomnia Severity Index (ISI), and Pittsburgh Sleep Quality Index
- Screening for other sleep disorders: RLS (sIRLS), sleep apnea (StopBANG), and circadian rhythm disorders (mCTQ)
- Screening for psychiatric comorbidities: depression (see depression for screening tools), anxiety (GAD-7), ADHD (ASRS), PTSD (PC-PTSD), and substance abuse

Cause of Insomnia

- **Medical conditions:** psychological comorbidities (above) + asthma, migraine, COPD, chronic pain, GERD, CHF, and thyroid dysfunction

Management of Insomnia

Goals of Therapy

- Improve daytime functionality and reduce impairment such as fatigue, dysphoria (this is the state of restless, frustration, and unhappiness)
- Promote restorative sleep despite disruptors

Non-Drug Measures

- Address cognitive barriers to sleep: dramatizing insomnia and putting so much emphasis on sleep, having unrealistic expectation, blaming insomnia for impaired daytime functionality
- Cognitive behavior therapy for insomnia (CBT-I) – may take 3-4 weeks for improvement
- Aerobic exercise
- Mediation, breathing techniques, light therapy, acupuncture, and weighted blankets
- **Sleep hygiene**
  - Bedroom: quiet, comfortable, and dark with a temperature of 16 – 20 degrees and use alarm not a clock in the bedroom. If using a clock, turn the clock face away
  - During the day: avoid daytime napping of more than 15-20 minutes and not later than 2:00pm, avoid stimulants after 7 pm, get 30-40 minutes of exercise daily
  - Before bedtime: avoid large meals (carbohydrate snack is OK) take warm bath, and plan a quiet period before lights out
  - Habits: keep regular sleep-awake cycle, use bedroom for intimacy and sleep, avoid reading/working while in bed, avoid looking at cell phone/TV/computer screen, and limit the time in bed to your nightly average. Educate the patient to go to bedroom only when sleep/drowsy and to not fight the inability to sleep

Drug Measures for Insomnia

Short-term insomnia: identify and address triggers + short term hypnotic and follow up within 2 weeks

- **Effective:** D/C therapy
- **Insomnia persists OR Chronic insomnia**

- If insomnia is because of sleep disorders → refer to a specialist/sleep center
- If insomnia is comorbidities-related → manage comorbidities
- **If primary insomnia is diagnosed**: assess + non-pharmacological measures (CBT-I is 1st choice) + hypnotic (1-2 weeks) → reassess sleep/daily functionality/ drug adverse reactions within 2 weeks
  - **Problems identified:** adjust doses and reassess
  - **Effective:** follow up in < 1 month
    - **Effective:** continue CBT-I and taper/stop the hypnotic
    - **Insomnia persists:** continue therapy

      Follow up every 3-6 months
- **If long term hypnotic is required:** continue CBT-I, use stable dose of hypnotic, and assess the adverse reactions, and the patient should return every 3-4 months for assessment and refills

When managing insomnia:

- Sleep hygiene is integral part of treatment
- Using a combination of non-pharmacological measures + hypnotics = more effect
- Sedatives should be taken PRN (up to 4 nights/week) using the lowest effective dose
- Hold hypnotic during low stress time
- All hypnotics cause some level of daytime drowsiness and may impair driving, but habituation is common
- When hypnotic is discontinued, the patient should expect 2-3 nights of poor sleep
- Shorten sleep time x 20 minutes 2 nights before hypnotic withdrawal and keep the shortened sleep time for 1 week.

Insomnia in Special Populations

Elderly

- Safe options: Doxepin (< 6mg/day) for maintenance insomnia, melatonin for initiation insomnia, and lemborexant (for up to 6 months)
- Trazodone and mirtazapine can be used
- BDZ and Z-drugs should be avoided, but if indicated, zolpidem and zaleplon can be used (they have short half-life) and low dose eszopiclone for up to 6 months
- Assess the benefits versus risk of discontinuing sedatives

Children

Melatonin (especially in children with autism and ADHD) and L-tryptophan. Clonidine can improve sleep in patients with ADHD. Other sedatives are not recommended

Pregnancy

- Insomnia is common in pregnancy
- CBT-I is the 1st choice in mild-moderate cases. If drug therapy is warranted (benefit > risk), zopiclone, eszopiclone, and trazodone can be used PRN in low dose. Avoid BDZ especially in the 1st trimester and near term. Switch long acting to short acting BDZ before labor. **AVOID** zolpidem during pregnancy. Doxepin and antidepressants are not studied but appear safe. Safety of Lemborexant, melatonin, L-tryptophan, and diphenhydramine is not established

Breastfeeding

- Short term use of intermittent low doses of short acting BDZ, zopiclone and zolpidem can be used during breastfeeding. Trazodone is unlikely to affect the breastfed infant.
- Safety of other agents is not established

Z-Sedatives (Z-Drugs)

- Include zopiclone, eszopiclone, zolpidem, and zaleplon
- **Mechanism:** BDZ agonists
- **Advantages over BDZ:** less rebound insomnia, dependence, cognitive impairment, and adverse reactions, and delayed tolerance
- Impair next day performance, ↓ coordination, and ↑ risk of falls
- They cause complex sleep behaviours with reported mortality (alcohol ↑ the risk)

Zaleplon: effective in initiation insomnia but requires compounding. It impairs driving for 8 hours

Suvorexant (Belsomra®) was discontinued for business reasons in 2020

Lemborexant (Dayvigo®)

- **Mechanism:** OX1R and OX2R antagonist (DORA= Dual orexin receptor antagonist)
- **Indication:** initiation and maintenance insomnia. It should be used when the patient has 7-8 hours of planned sleep
- **Contraindications:** narcolepsy. It should not be used with moderate/strong 3A4 inhibitors/inducers, alcohol, or other CNS depressants
- **Adverse reactions:** drowsiness (most common), headache, nightmares, fatigue, memory loss, sleep paralysis. gait problems, nasopharyngitis, complex sleep behaviors (e.g., sleep driving, cooking, eating, etc. along with amnesia – D/C immediately), hypnagogic/hypnopompic hallucinations, driving/functioning impairment (dose-dependant, patient feels fully awake, allow 7 hours before driving/operating machinery), depression (worsens depression and suicidality)
- Minimal daytime drowsiness, cognitive impairment, driving impairment, and abuse potential
- **Onset:** 30 minutes but it can be delayed if taken with or shortly after food

- **Dose:** 5 – 10 mg at bedtime +/- food (but consider the delayed onset). 5mg is the maximum dose in hepatic impairment and weak 3A4 inhibitors

Other hypnotics

- Doxepin in small dose (3 – 6 mg) is approved for maintenance insomnia. It does not cause rebound insomnia, nor dependence and it causes minimal daytime drowsiness.
- Trazodone: widely used for primary maintenance insomnia. Commonly used in elderly (25-100mg)
- Cannabis: has insufficient evidence for routine use but can be used in patients with resistant insomnia + other comorbidities (e.g., neuropathic pain or anxiety)
- Valerian

ADHD

No updates

ACUTE AGITATION

No updates

EATING DISORDERS

No updates

EPILEPSY

NEW CONTENTS

Causes of recurrence while on antiepileptic therapy

- Non-adherence (common problem)
- Sleep deprivation
- Intercurrent diseases (e.g., acute illnesses)
- Abuse of alcohol or other drugs
- Drug withdrawal
- Acute structural brain injury

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Females of childbearing age	Antiepileptic drugs ↓ effect of combined contraceptives (minimal with gabapentin, pregabalin, lacosamide, and levetiracetam). ▪ **Most affected:** CHC (oral, vaginal, patches), POP, and progestin implants. ▪ **Least affected:** depo-MPA (1st choice), IUD and IUS and barriers (least effective). May consider a combination of CHC that provide ≥ 50-60 mcg of EE ▪ Barriers can be added to ↑ the contraceptive effect + protect against STI but they are unreliable to replace hormonal contraceptives ▪ CHC ↓ serum level of lamotrigine by 50% (monitor lamotrigine level)

NEW CONTENTS

Common Adverse Reactions

- **Dose-dependent:** nausea, sedation, dizziness, ataxia, fatigue, and cognitive impairment. Dose reduction helps. If the adverse reaction happens 30-60 minutes post-dose (i.e., peak-related following immediate release dose), switch to sustained release or more frequent administration of lower dose
- **Rash:** onset is 6 weeks but can happen at any time. All agents cause rash but most with phenytoin, CBZ (risk ↑ among Chinese and Japanese), and lamotrigine. Potential cross reaction among AED. D/C AED and switch to a drug with low potential of different chemical
- **Osteoporosis:** with valproic acid and enzyme-inducing AED. Monitor BMD + Ca/vit-D supplement

PARKINSON DISEASE

No updates

HEADACHES

NEW AND REVISED CONTENTS

Assessment

- No diagnostic tests for primary migraine
- The presence of ≥ 2 of the following during migraine suggest migraine: nausea, being troubled by light, or limited ability to work/study for at least 1 day
- CT/MRI, lumbar puncture, ESR/CRP, and neurological assessment can be used to rule-out other causes.

Clinical Presentation, Classification, and Diagnosis

Type	Attacks meet criteria	Average Migraine days/month	Duration	≥ 2 of the following				Other symptoms	
				Location	Pulsating	Severity	Daily activities	Nausea/ Vomiting	Photo/ Phonophobia
Tension	≥ 10	**Infrequent Episodic:** < 1 (<12/year) **Frequent Episodic:** 1–14 x > 3 months	½ - 7 hours	Bilateral	No (pressing or tightening)	Mild-moderate	Do not aggravate	NO	Only one
		Chronic: ≥ 15 x > 3 months	Hours – days or unremitting					Photo or phono-phobia or mild (not severe) nausea, and no vomiting	
Migraine without aura	≥ 5	**Episodic:** < 15 **Chronic:** > 15 x > 3 months	4 – 72 hours	Unilateral	Yes	Moderate – severe	Aggravate	≥ 1 of the following: ▪ Nausea **and/or** vomiting **and/or** ▪ Photo + phonophobia	

Migraine with aura	≥ 2	**Episodic:** < 15 **Chronic:** > 15 for > 3 months (requires 5 episodes)	≥ 1 of the following reversible aura symptoms: **visual** (**positive:** scintillations = flashing spots/lines or **negative:** loss of vision), **motor** (1-sided or facial weakness), **sensory** (**positive:** pricking feeling or **negative:** numbness), **speech** (dysarthria, aphasia), and **neurological** (tinnitus, vertigo, ataxia, diplopia, ↓ consciousness, and hypoacusis) **PLUS The aura symptom(s) must meet ≥ 3 of the following criteria:** aura is accompanied by the migraine or followed by migraine within 60 minutes, at least 1 aura symptom spreads gradually over > 5 min, each symptom lasts 5 – 60 minutes, symptoms occur in succession, at least 1 aura symptom is unilateral, and at least 1 aura symptom is positive
Med overuse migraine		≥ 15 days	The patient has pre-existing headache disorder + use ≥ 1 abortive therapy for ≥ 10 days/month (in case of opioids, triptans, or ergots) or for ≥ 15 days/month (in case of conventional analgesics) for 3 months. Other non-diagnostic symptoms include the presence of headache upon waking up, headache is severe and interferes with daily life, and abortive therapy becomes less effective
Cluster Migraine	≥ 5 occur 1 q2d or up to 8/day	The attacks meet the following criteria: ▪ Severe or very severe ▪ Location: Unilateral – orbital, supraorbital or temporal ▪ Lasting 15 – 180 minutes if not treated ▪ ≥ 1 of the following: hyperemia, lacrimation, nasal rhinorrhea or congestion, eyelid edema, facial sweating, miosis or ptosis, restless or agitation **Episodic:** ≥ 2 clusters last 7 – 365 days and separated by > 1 month of pain-free period **Chronic:** Reoccur for > 1 year without remission or remission lasting < 1 month	

- Headache Classification Committee of the International Headache Society (IHS) The International Classification of Headache Disorders, 3rd edition. (2018). *Cephalalgia: an international journal of headache*, *38*(1), 1–211.
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Red Flags of Migraine (SNNOOP10)

S	Systemic symptoms: fever, weight loss, rash, joint pain, etc.
N	Neoplasm
N	Neurological symptoms (but not aura): ↓ consciousness, altered vision, neck stiffness, etc.
O	Onset: sudden, maximal intensity within seconds to minutes – refer to ER
O	Older age: > 50 years of age
P (10)	Pattern: changing frequency or severity Postural headache: headache occurs when sitting or standing and goes away after lying down Precipitated by cough or sneezing Papilledema Progressive or atypical presentation of headache Pregnancy and postpartum: new onset of severe headache occurs during pregnancy and postpartum Pain of the eye with symptoms such as hyperemia, tearing, nasal congestions, facia pain, etc. Post-traumatic migraine Pathology of immune system Pain killer overuse

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- Headache Classification Committee of the International Headache Society (IHS) The International Classification of Headache Disorders, 3rd edition. (2018). *Cephalalgia: an international journal of headache*, *38*(1), 1–211.

Management of Acute Attacks

Goals of Therapy

- Alleviate (or abolish) pain,
- Develop coping strategy
- Improve QOL, and restore functionality
- Prevent drug adverse reactions including medication overuse headache (by minimizing the need for repeat doses or rescue medications)

Non-pharmacological measures + abortive therapy
Standard abortive therapy: Triptan +/- Simple analgesic +/- antiemetic drug

Therapeutic Tips

- **Target:** pain-free (ideal) or relief at 2 hours with 24-hour relief
- Start abortive therapy as soon as possible
- Mild attacks may not require treatment
- Simple analgesics (e.g., acetaminophen, naproxen, ibuprofen, ASA, etc.) are first line in mild acute attacks. NSAID are more effective, but acetaminophen is safer. If they fail, consider triptans
- Combination of triptan + NSAID is more effective but more adverse reactions
- Educate the patient to select the abortive therapy based on the "traffic light system"
- Antiemetics ↑ effect of other agents even in absence of nausea or vomiting.
- Limit the use of simple analgesics to < 15 days/month and triptans to < 10 days/month
- **Use of triptans:**
  - Dosing: either low dose and may repeat after 2 hours or high dose once
  - Trial: try a triptan for at least 2 attacks (re-dose and/or ↑ dose) → If fail, try a different triptan. Try 3 agents before considering other agents
  - If there is no relief at 2-hour post triptan or cannot use triptan for recurrence: add NSAID
- May consider non-oral route (e.g., NSAID suppository or SC/nasal triptan) in patients with nausea/ vomiting
- Ergots are associated with ↑ ADR and ↓ effect compared to triptans. Can be used as alternative but generally unfavorable
- If standard therapy fails or contraindicated: Consider: opioids or tramadol +/- butorphanol/ butalbital (last resort because of tolerance/dependence). Consider prophylactic therapy
- **In acute care settings:** IM ketorolac, IV/IM meperidine, or IV DHE
- **Antiemetics:** metoclopramide IV/SC/PO (best evidence), prochlorperazine PR/PO, dimenhydrinate PO/PR, chlorpromazine PO/IV, domperidone PO/PR, dexamethasone or IV, haloperidol IV
- **Status migraines:** DHE IV

Prophylactic Therapy of Migraine

Goals of Therapy

- Reduce: attack severity/frequency/duration/impact on QOL, related stress and psychological burden, and the use of acute treatment/cost of therapy
- Improve: QOL, functionality

Candidates

- Migraine that impacts QOL OR
- ≥ 4 headache days/month (≥ 6 headache days/month – RxFiles) OR
- Abortive therapy is contraindicated, not effective, associated with adverse reactions, or used frequently

Target: ≥ 50% ↓ in number of attacks or days with migraine, ↓ severity/duration of attacks, improved QOL and functionality, and better response to abortive therapy.

Try one agent and ↑ the dose every 1 – 2 weeks and continue for 2 – 3 months after dose titration

- If effective → continue x 6 – 12 months, then taper very slowly → if migraine rebounds (no. of attacks or severity) ↑ dose to the effective dose. Indefinite treatment may be required
- If partially effective: add another 1st line agent
- If ineffective → D/C and try a different drug
- If the patient fails ≥ 2 agents: CGRP mABs or onabotulinumtoxinA

- **Drug classes:**
  - Beta blockers: all but **propranolol and metoprolol are 1st choice.** Useful in hypertension
  - CCB: verapamil (best evidence in cluster headache – useful in hypertension), and flunarizine (causes weight gain and depression)
  - TCAD: **amitriptyline is 1st choice;** nortriptyline is alternative if amitriptyline is not tolerated. Useful in depression, anxiety, or insomnia
  - SNRI: venlafaxine and duloxetine. Useful in depression, or anxiety
  - AED: valproic/divalproex, **topiramate (1st choice.** Useful in obese**)**, and gabapentin (3rd line – has insufficient evidence). Useful in epilepsy
  - Calcitonin gene-related peptide monoclonal antibodies (CGRP mABs – erenumab, galcanezumab, fremanezumab and eptinezumab. All 3rd line)
  - ACEI/ARB: lisinopril, candesartan, and telmisartan – all 3rd line. Useful in hypertension
  - NMDA: memantine. Used after failing other therapies
  - Serotonin modulators: triptans (used for menstruation-associated migraine. Used 5 – 7 day/cycle starting 2 days before predicted onset of migraine), and pizotifen (causes weight gain and sedation)
  - Cyproheptadine: antihistamine that is typically used in migraine prevention in children
  - Long cycle birth control to prevent menstruation-associated migraine
  - Supplements (all 3rd line): Melatonin (useful in insomnia), butterbur (has the best evidence, burping is the most common adverse reaction. May cause hepatotoxicity), riboflavin, feverfew (questionable evidence. Withdrawal symptoms include severe migraine and insomnia), coenzyme Q10, and Mg citrate

NEUROPATHIC PAIN

No updates

PAIN MANAGEMENT

No updates

MULTIPLE SCLEROSIS

No updates

RESTLESS LEG SYNDROME

No updates

MISCELLANEOUS TOPICS

No updates

DRUG INFORMATION

I developed the following new table for all antidepressants. The table was not recorded, but I believe it is useful and easier to follow

Endocrine Updates

DIABETES

NEW CONTENTS

Symptoms of Diabetes

- **Metabolic symptoms:** polyuria, polydipsia, weight loss
- **Complication symptoms:** DKA, or presence of micro/macrovascular complications
- **Non-specific symptoms:** fatigue, weight changes, lack of energy, recurrent vaginal candidiasis.

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Risk Factors for Diabetes (new ones to add)

- Low socioeconomic status
- Family history of NIDDM
- **Drugs:** isoniazid, 5-flurouracil, systemic decongestants, estrogen, GnRH agonists, pentamidine, mammalian target of rapamycin (mTOR) inhibitors (e.g., sirolimus), and as a complication of parenteral nutrition and TPN

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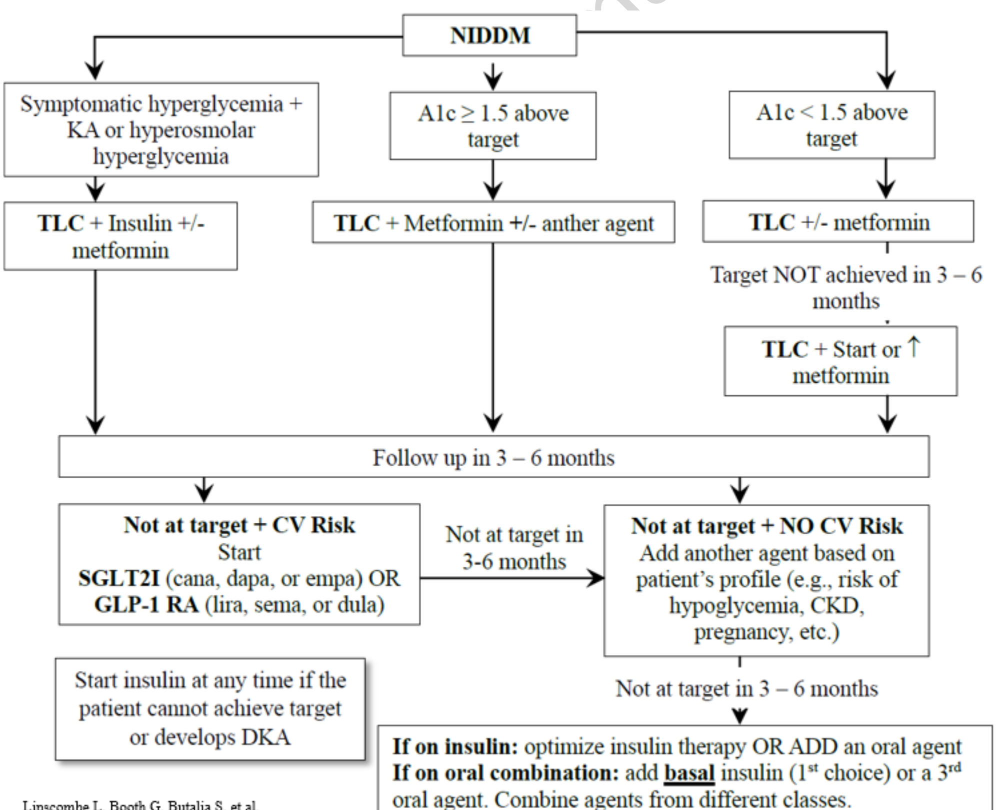

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Management of hypoglycemia

- **Mild – Moderate**:
  - **ADULTS**: 15g of sugar (**20 g if SEVERE but conscious**) → check BG in 15 minutes → if < 4, give 15 g of sugar (for all cases)
  - **CHILDREN:** Children < 15 Kg = 5g sugar, 15-30 Kg = 10 g and > 30 Kg = 15 g
- **Unconscious (Severe) + IV access:** Give 10 - 25 g of glucose IV (20 – 50 ml D**50%**) over 1 – 3 minutes
- **Unconscious (Severe) + NO IV access:**
  - **ADULTS:** 1 mg glucagon SC or IM or 3mg nasal powder (if the patient is ≥ 4 years). Patients and caregivers prefer nasal powder. BG ↑ in 15-30 minutes.
  - **CHILDREN:** Glucagon in a dose of 15-30 mcg/Kg to a maximum of 1 mg
- **15g of sugar (usually ↑ BG by 2 mmol/L in 20 minutes) = any of the following:** 15g glucose in the form of glucose tablets, 3 teaspoonful or 3 packets of table sugar dissolved in water, 150 mL (3/4 cup) of juice or regular soft drink, 6 LifeSavers (2.5 g carbohydrate/each), or 15 mL (1 tablespoonful) of honey
- **Once resolved** → All patients should receive a regular meal (or snack if regular meal is due in > 1 h) that contains carbohydrates and proteins to prevent recurrence
- Educate on hypoglycemia and assess for unawareness
- Consider ↑ target for up to 3 months following hypoglycemia (particularly if severe)
- Glucagon is NOT effective in malnourished patients or in alcohol-induced hypoglycemia. Glucose or dextrose should be used in these patients

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NEW CONTENTS

Different Types of Monitoring

Monitoring modalities can be classified according to the fluid being used for monitoring into

- **Capillary Blood Glucose monitoring (CBG):** monitoring the capillary blood glucose using a blood sample obtained by pricking the finger or an alternative testing site (e.g., forearm – depending on the device). It is used to be called SMBG
- **Interstitial fluid monitoring:** ongoing monitoring the blood glucose level in the interstitial fluid (delayed level compared to blood glucose). There are different modalities
  - **Intermittently scanned continuous glucose monitoring (isCGM) – used to be called Flash Monitoring:** The patient scans a sensing device from time to time. Device example: **Freestyle Libre** – sensor (used for 14 days), and a scanner (can be a mobile application or Libre Reader). Each scan gives the glucose level in the last 8 hours. No alarm functions. isCGM was found to decrease the time spent in hypoglycemia, and increase the time BG in range
  - **Real-time continuous glucose monitoring (rtCGM) – used to be called continuous glucose monitoring:** the sensing device is connected to a transmitter that continuously transmit the reading to a device. The patient can view the readings at any time without scanning. Device example: **Dexcom G6** – sensor (used for up to 10 days), transmitter (sends data to a mobile application), and a mobile application or a receiver. rtCGM improves A1c, ↓ the time in hypoglycemia, and improves QOL and treatment satisfaction
  - **Masked continuous glucose monitoring (mCGM) – used to be called professional or blinded CGM:** the sensing device store the data to be retrieved at any time. It is used by professionals. The sensor can be used for up to 5 days. The transmitter stores the data, and the patient does not get any readings and thus cannot adjust insulin doses based on the readings. mCGM improves A1c in adults with type 2 diabetes and in pregnant women with type-1 or type-2 DM

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REGULAR INSULIN

Entuzity is available in 500 units/mL and should be reserved for patients with severe insulin resistance (those require > 200 units of basal/bolus per day). Entuzity is not biosimilar to regular insulin and has duration like NPH (↑ concentration delays the onset and prolong the duration)

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RAPID ACTING INSULIN

Rapid acting insulins are suitable for insulin pump except the non-standardized formula

Fiasp® is not biosimilar to NovoRapid (contains arginine and nicotinamide). Fiasp is faster acting and administered 5 minutes before and up to 20 minutes after starting the meal. Fiasp should not be mixed with other insulins or used in pump. **Admelog®** is biosimilar to Humalog

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BASAL INSULIN ANALOGUES

- (AM or HS but HS preferred) or twice (less common – used if action does not last 24 hours or patient requires large dose).
- **Basaglar®** is biosimilar to Lantus and they are not interchangeable.
- **Toujeo** is reserved for patients who requires > 20 units of Lantus/day. Dose adjustment is every 3-4 days (Lantus/Basaglar/Levemir every 1-2 days)

HYPOTHYROIDISM

No Updates

HYPETHYROIDISM

No Updates

CONTRACEPTIVES

NEW CONTENTS

Etonogestrel Implant (Nexplanon®)

- **Indication:** prevention of pregnancy for 3 years. Failure rate is 0.05%. It is preferred in lactating individuals, postabortion, postpartum, or if estrogen is contraindicated
- **Contraindications:** Similar to POP + current/history of thrombosis
- **Dose:** one implant provides protection x up to 3 years and it should be removed after NO more than 3 years. The implant can be removed at anytime upon patient's request.
- **Reversibility:** after removal, blood level ↓ in 1 week with pregnancies observed as early as 7 to 14 days after removal
- It gets less effective in overweight women, and more frequent dosing may be required
- It was not studied in patients with liver, and renal impairment and obese patients
- Storage: between 2 – 30 degrees

- **Administration:**
  - Subdermal implant inserted under the skin at the inner side of the non-dominant upper arm
  - Insertion and removal must be performed by trained health professional
  - Immediately after insertion, the implant must be palpated by the inserting professional and the patient. The patient must occasionally palpate the site to confirm the presence of the implant
  - Insertion complications:
    - Improper insertion can cause complications such as pain, paresthesia, bleeding, hematoma, scarring or infection when inserted or removed
    - Deeper insertion: it may not be palpable/localized, and removal can be difficult. Deeper insertion causes neural damage (e.g., paraesthesia) or vascular damage
    - Migration to the blood vessels of the arm and the pulmonary artery: If the implant is not palpable, it should be located, and removed as soon as possible. X-ray or ultrasound of the arm or the chest may be required to locate the implant

Initiating therapy

- **Female is not taking contraceptive:** insert between day 1-5 of the cycle even if still bleeding
- **Switching from CHC pills:** insert the day after the last active tablet (preferred) or on the day when the next cycle of drug should start
- **If on progestin only method:** insert on the day the next injection is due (MPA), the same day IUS is removed, within 24 hours after taking the any POP tablet (i.e., at any time in case of POP)
- **Following abortion:**
  - **First trimester abortion:** within 5 days following abortion
  - **Second trimester:** insert between 21 to 28 days following abortion

If inserted as above, no back-up method is required. Otherwise, use backup method x 7 days

- **Postpartum:** can be inserted immediately (RxTx), but monograph recommendations are
  - **Breastfeeding:** insert after the 4th week postpartum + back-up method x 7 days
  - **No breastfeeding:** insert between 21 to 28 days postpartum. back-up method x 7 days is required if inserted beyond day 28 postpartum

Adverse Reaction: Insertion site pain, menstrual irregularity (spotting, BTB, and amenorrhea. Unexplained bleeding should be investigated), emotional lability, weight gain, headache, depression (monitor), acne, breast tenderness, abdominal pain, flatulence, nausea, decreased libido, dizziness, nervousness, dysmenorrhea, insulin resistance (monitor – likely insignificant), ↑ LDL, and thrombosis (patient must report symptoms suggesting DVT, stroke, and MI – remove the implant in case of long-term immobility. Discourage the use in patients with uncontrolled HTN and monitor HTN during therapy)

Drug interactions:

- Efavirenz, CBZ, phenytoin, barbiturates, rifampin, topiramate, St. John's wort: ↓ etongestrel level (most significant with rifampin – back-up method is required)
- Azoles antifungals, GFJ: ↑ etongestrel level
- Protease inhibitors: ↑ or ↓ etongestrel level depending on the agent

MENOPAUSE

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Definitions:

- **Perimenopause (menopause transition):** cessation of the ovary function (ovulation, and release of sex hormones)
- **Menopause:** amenorrhea and lack of estrogen x 1 year. The average age is 51 years. Menopause can be induced.
- **Early menopause:** menopause in females < 45 years of age
- **Premature menopause:** menopause in females < 40 years of age

Symptoms associated with menopause

- **Vasomotor symptoms:** most common symptoms are hot flashes and night sweats (start 2 years before and peak 2 years after the last menses and disappear in about 7 - 8 years), and migraine. Will improve over time.
- **Genitourinary syndrome of menopause (GSM):** a combination of
  - **Vaginal symptoms:** vaginitis, itching, dryness, and increased risk of vaginal infections
  - **Urinary symptoms:** urinary incontinence and urgency, dysuria, and increased risk of UTI
- **Skeletal:** ↑ risk of osteoporosis, arthralgia
- **CNS:** mood swings, fatigue, irritability, headache, and sleep disturbance.
- **Sexual:** dyspareunia, and ↓ libido
- **Others:** dry skin and tachycardia

Non-Drug measures

- Smoking cessation: ↓ frequency and severity of hot flashes and night sweats
- Weight loss (↑ vegetables/fruits/whole grains, and ↓ fat): faster elimination of VMS
- Cooling techniques: avowing heavy clothes, dressing in layers, and using fan/air condition
- Avoid triggers: alcohol, caffeine, and spicy foods and rinks,
- Exercise (≥ 150 minutes/week) and relaxation therapy (stretching exercise, yoga and paced respiration) improve QOL
- Clinical hypnosis reduces vasomotor symptoms
- Cognitive behavior therapy (CBT)

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- **Uses: for VMS** when HRT is contraindicated (e.g., in patients with history of estrogen-sensitive breast cancer), as an alternative if HRT is undesirable, or in combination with HRT
- **Venlafaxine or (es)citalopram:** 1st choice in hot flashes PLUS mood changes and/or anxiety
- **Other SSRI** and **SNRI** can be used off label for hot flashes. They take 3 – 4 weeks for full effect. But **paroxetine** may complicate the condition because of weight gain. Avoid **fluoxetine** and **paroxetine** in breast cancer survivors on tamoxifen
- **Gabapentin**, **pregabalin**, and **clonidine** (try x 2-4 weeks and D/C if no benefit) are effective for hot flashes
- **Oxybutynin** ↓ VMS
- **Phenobarbital** + **ergotamine** + **belladonna** combination (Bellergal-S®)

NEW CONTENTS

Tibolone (Tibella®)

- **Mechanism:** tibolone is a norethynodrel analogue. It is a prodrug that is converted into three active metabolites which produce weak estrogenic, progestogenic, and androgenic effects
- **Indications:** short term management of VMS in females with intact uterus
- **Onset:** few weeks until the patient feels improvement in hot flashes
- **Contraindications:** like hormone replacement therapy
- **Adverse reactions:** hair growth, acne, breast tenderness, ↑wt; Breakthrough bleeding/spotting, breast tenderness, abdominal pain (bloating, water retention), ↑ appetite and weight gain, nausea, upset stomach, fatigue, acne, and hair growth
- **Drug interactions:** serious interactions 1) ↑ fibrinolytic effect and ↑ bleeding in patients taking anticoagulants (particularly warfarin) and 2) St. John's wort ↓ estrogenic and progestongenic effect leading to change in uterine bleeding profile
- **Dose:** 2.5mg daily +/- food. It should be used for 6 months, then re-assess

OESITY

No Updates

MISCELLANEOUS TOPICS

No Updates

Respiratory Updates

ASTHMA IN ADULTS

Page 166

- PRN SABA as monotherapy reliever should only be considered in patients with very mild asthma + low risk of exacerbation. Candidates:
    - SABA use < 2x/week and ≤ 2 canisters/year
    - Nonsmokers
    - No history of exacerbation requiring systemic steroid or hospitalization

1	2	3	4	5
Preferred: PRN ICS + Formoterol **Alternative:** low dose ICS (given along with SABA)	**Preferred:** DAILY Low dose ICS **OR** PRN ICS + Formoterol **Alternative:** low dose ICS (**OR** LTRA) taken when SABA is used	**Preferred:** Low dose ICS + LABA **Alternative:** Medium dose ICS **OR** Low dose ICS + LTRA +/- sublingual immunotherapy (SLIT)	**Preferred:** Medium dose ICS + LABA **Alternative:** High dose ICS **OR** ADD-LAMA (triple therapy device or tiotropium mist) **OR** ADD-ON LTRA. Consider add on SLIT	**Preferred:** High dose ICS + LABA +/- tiotropium mist, omalizumab, or IL-5 or IL-4 antagonist **Alternative:** Low dose oral steroid

Reliever:
1st choice: PRN ICS + Formoterol
Alternative: PRN SABA (salbutamol or terbutaline)

Triple Therapy Devices
- Enerzair Breezhaler = mometasone +indacaterol + glycopyronium
- Trelegy Ellipta = Fluticasone + vilanterol + umeclidinium
- Advantages: convenience, adherence, ↓ cost, and ensure ICS use. Disadvantage: loss of dosing flexibility

NEW CONTENTS

Adolescents are at higher risk of
- Poor adherence: risk factors include forgetfulness, fear of stigma, cost of medications, substance use, comorbidities, and the added responsibility to manage asthma. Reviewing adherence and risk factors are important
- Poor control of asthma
- ↑risk of morbidity and mortality

Concomitant asthma + COPD
- Ensure adequate ICS + sufficient bronchodilation
- Add other therapies according to the patient's symptoms/needs (e.g., eosinophile level, lung functions, risk of exacerbation, etc.)

ASTHMA IN CHILDREN

No Updates

COPD

No Updates

SMOKING CESSATION

Page 177

- E-cigarettes or Electronic Delivery System (EDS) – inhaling the vapor of nicotine + PEG + other ingredients, may have some risks (FDA report), and is not regulated. Reports links E-cigarettes to **e**-cigarette, or **v**aping, product use-**a**ssociated **l**ung **i**njury (EVALI) due to vitamin E-acetate– generally should not be recommended as smoking cessation aid.
- Symptoms are NOT specific: SOB, fever, cough, chest pain, palpitation, dizziness, vomiting, and diarrhea

INFLUENZA

No Updates

VIRAL RHINITIS

No Updates

ALLERGIC RHINITIS

Page 192

Non-Allergic Rhinitis

Causes of Non-Allergic Rhinitis
- URTI
- Alcohol intake
- Pregnancy and during menstruation
- Vasomotor rhinitis: chronic (runners or wet/dry): cold/dry air, perfumes, air pollutants, temperature fluctuation, hot and spicy foods and beverages, intense emotions, bright light, smoking, and smog.
- Nasal causes: nasal polyps, deviated septum, and nasal passage atrophy,
- Medical disorders: hypothyroidism, nonallergic rhinitis with eosinophilia syndrome (NARES)
- Drugs: Beta blockers, CCB, ASA, ACEI, hydralazine, diuretics, NSAID, sympatholytic drugs (alpha-1 blockers and alpha-2 agonists), gabapentin, cocaine, antipsychotics, PDE5-I, and oral contraceptives. Prolonged use of topical decongestants and sympathomimetic drugs (e.g., amphetamine) causes rhinitis medicamentosa

CROUP

No Updates

COUGH

No Updates

Miscellaneous Topics Updates

HYPOKALEMIA

No updates

HYPERKALEMIA

No Updates

HYPERCALCEMIA

No Updates

PALLIATIVE CARE

Please see the new chapter and video on the portal under "Cancer Care"

CHRONIC KIDNEY DISEASE

<u>Page 257</u>

See the revised table for the management of CKD comorbidities

<table>
<tr><th colspan="2">Management of Comorbidities & Complications</th></tr>
<tr><td>Hypertension</td><td>

Management: ACEI or ARB particularly when there is proteinuria (slow CKD progression, ↓ free radicals, ↓ glomerular pressure and proteinuria) → ↑ dose and/or add a thiazide diuretic (if diabetic, adding CCB is preferred) → add other agents. Allow 4 – 6 weeks between steps.
ACEI and ARB should be titrated to target dose even in normotensive patients (monitor BP and kidney function)
If BP is < 130/80 before initiating therapy, monitor for hypotension
Monitor eGFR and K^+ at baseline and 2 weeks after initiating/ increasing ACE/ARB dose

D/C ACEI/ARB if eGFR decreased by >50%. ↓ dose if eGFR drops by 30 – 50% and no dosage change but monitor is eGFR drops by < 30%
If K+ is 5-6mmol/L, follow dietary restrictions. If K+ is 6-6.5mmol/L, add loop diuretic +/- K-exchange resin
Repeat monitoring

Target blood pressure (if safely achievable)
<table>
<tr><td>Patients' Group</td><td>Target BP</td></tr>
<tr><td>Diabetic + proteinuria</td><td>≤ 130/80</td></tr>
<tr><td>Diabetic + no proteinuria</td><td>≤ 140/90</td></tr>
<tr><td>Nondiabetics 50 – 75 years + FRS ≥ 15% - no stroke + no advanced CKD</td><td>SBP < 120</td></tr>
<tr><td>Nondiabetics + polycystic kidney disease</td><td>SBP < 110</td></tr>
<tr><td>Nondiabetics + frail, residents of LTC, stroke, life expectancy < 3 years, taking > 5 chronic medications, and standing SBP < 110</td><td>SBP < 140</td></tr>
</table>
</td></tr>
</table>

Diabetes Type II	▪ CKD ↑ risk of hypoglycemia (monitor, ↓ dose of insulin and hypoglycemic agents).
	▪ Metformin + SGLT-2 inhibitors are 1st choice
	▪ D/C metformin when CrCl/eGFR < 30ml/min or increased risk of lactic acidosis
	▪ **SGLT-2I:**
	  o ↓ progression of CKD, need for dialysis, and mortality.
	  o Consider SGLT-2I for diabetic patients who receive ACEI/ARB + ACR > 22mg/mmol (↓ mortality, HF, and slow CKD) and for nondiabetic CKD with ACR > 100mg/mmol.
	  o DO NOT initiate if eGFR is < 30ml/min
	  o Ensure euvolemia and normal/elevated BP before initiating therapy
	  o If eGFR ↓ by > 20-25%, hold, and assess for ARF. Monitor for euglycemic DKA
	  o D/C SGLT-2I when the patient requires dialysis.
	  o SGLT-2I ↑ risk euglycemic DKA in patients type I DM – avoid
	▪ Other agents can be added as needed based on eGFR, comorbidities, and patients' preference. GLP-1 analogues are preferred. Monitor and adjust doses as needed
	▪ **Stage 4 CKD:** consider insulin, gliclazide and glimepiride (avoid other SU), gliptins (saxa, sita, lina, and alogliptin), and TZD
	▪ **Stage 5 CKD:** consider insulin, gliptins (except saxagliptin), and TZD can be used

↑ Cardiovascular Risk	▪ Patients with CKD are at high risk for cardiovascular events ▪ Control comorbidities e.g., hypertension and diabetes ▪ Risk is associated with the amount of protein in urine > decline in eGFR ▪ Start statin +/- ezetimibe, regardless of LDL, in CKD patients ≥ 50 years and CKD patients ≥ 18 years with diabetes, CAD, stroke, and FRS > 10%. Fibrates are not recommended ▪ ASA has questionable effect in primary prevention. Use when needed (e.g., post-MI)
Hyperkalemia	If K^+ > 5 mmol/L → D/C potassium supplements → restrict dietary potassium → add loop diuretic (keep ACEI, ARB and MRA if used to maintain kidney function)
Metabolic acidosis	Sodium bicarbonate or Shohl's solution (citric acid + sodium citrate) or K-Shohl's solution (citric acid + K- citrate)
Hypercalcemia	As discussed before

Hyperphosphatemia	▪ ↓ phosphate-rich diet is the 1st choice (seafood, seeds, pork, nuts, beef, low fat dairy products, soya foods, beans, and lentils)
	▪ If phosphate remains elevated AND
	  o **Calcium level is normal:** calcium-containing phosphate binder (ex. calcium carbonate or citrate) is the 1st choice. Monitor Ca level and ↓ dose if necessary → if hypercalcemia develops then see below
	  o **Hypercalcemia:** non-calcium phosphate binder: **sevelamer (the most common ADR is nausea and vomiting**. Carbonate is preferred over hydrochloride because it neutralizes metabolic acidosis), **lanthanum**, and sucroferric oxyhydroxide
	▪ Vit. D analogues (calcitriol and alfacalcidol) – ↑ Ca level **without binding to phosphate**

Immunization	▪ Hepatitis B vaccine (to all patients. Increased risk of hepatitis B due to multiple IV penetration. May require higher dose and follow up tests to ensure immunity) ▪ Other vaccines: vaccines for tetanus, pneumococcal, influenza, varicella-zoster virus, Staphylococcus aureus, human papillomavirus, and rabies.

Vitamins & supplements in CKD (should be recommended by a specialist)

- Replavite® is a combination of vitamins B1 (thiamine), B2 (riboflavin), B3 (nicotinamide), B6 (pyridoxine), B7 (biotin), B12 (cyano-, hydroxy-, or methyl cobalamin), folic acid and vitamin C (100mg). It is recommended in patients with CKD who have poor eating habits (nausea, vomiting, eating less than 50% of meals, consumes < 1,500 kcal/day or has undesirable weight loss) or on hemodialysis to replenish the loss of water-soluble vitamins.
- Activated vitamin D (1,25 hydroxy) is also used. Note that vitamin D analogues, such as calcitriol, are NOT supplements but used for specific indications such as hypocalcemia and vitamin D-resistant rickets.
- Iron: If Hg < 100g/L or using erythropoietin. If PO fails, consider IV. May try PO in hemodialysis

BENIGN PROSTATIC HYPERPLASIA

No Updates

INCONTINENCE IN ADULTS

Page 261

ADD – a new anticholinergic drug for the management of urge incontinence – propiverine

INCONTINENCE IN CHILDREN

Definition: According to DSM-V

- **Enuresis in children:** bedwetting or wetting during sleep (e.g., nap time), more than twice weekly beyond the age of 5 years for girls and 6 years for boys. Subtypes of enuresis:
  - Volume-dependant: nocturnal polyuria due to impaired ADH secretion at night
  - Detrusor-dependant: associated with daytime frequency, urgency, or incontinence
  - Monosymptomatic: no other neurological, urological, or medical comorbidities
  - Non-monosymptomatic: bedwetting + daytime frequency, urgency, or incontinence
- **Incontinence:** repeated daytime or nighttime voiding of urine in the bed or clothes at least twice per week for at least 3 consecutive months, in a child who is at least 5 years of age.

Assessment

- Urine analysis and culture (treat if UTI is evident)
- If the child was previously dry in the last 6 months, investigate for a cause (2ry bedwetting) such as diabetes (mellitus or insipidus), UTI, sleefp apnea, psychological problems, or dysfunctional voiding.

Non-Drug Measures

- Keep child dignity (no punishment, no humiliation)
- Ensure access to bathroom
- Bladder training: regular voiding, and pelvic floor muscle exercise
- Stimulation therapy (e.g., laser acupuncture, and magnetic stimulation) ↓ symptoms and relapse
- Limiting milk intake ↓ risk of hypercalcemia and improve the symptoms
- Other therapies such as homeopathy and hypnosis lack evidence

Measures for daytime incontinence

Non-delayed voiding

Measures for enuresis

- Enuresis alarm (delayed onset) – best used in children > 7 years. Low relapse rate when used with dry bed training (below) or overlearning (the child challenges the ability to awaken by drinking 200 mL of fluids 1 hours before bedtime while using the alarm). It is used as a firs line in primary enuresis
- Avoid fluid intake within 2 hours before bedtime
- Dry bed training: wake up the child every night at the same time to develop the habit of waking up when needed

Drug Management

Management Approach

- Daytime incontinence: non-drug measures and oxybutynin or tolterodine
- Enuresis: non-drug measures + enuresis alarm or desmopressin → assess the response:
    - No-response: <50% reduction → ↑ desmopressin dose or combine desmopressin + enuresis alarm → no response continues = refractory case: ensure proper use of enuresis alarm + investigate other causes (e.g., social issues, overactive bladder, sleep apnea, etc.) → start a combination of desmopressin + anticholinergic drugs (↓ detrusor hyperactivity. Oxybutynin is the 1st choice. If oxybutynin is not tolerated, alternatives include tolterodine, solifenacin, or propiverine)
    - Partial response: 50 to 99% reduction
    - Complete response: 100% reduction → monitor long term success
        - No relapse at 6 months: continued success
        - No relapse at 2 years: complete success
        - More than 1 symptom recurrence/mont: Relapse → combine enuresis alarm + desmopressin
- Amitriptyline, desipramine and imipramine ↓ wet nights but have a narrow therapeutic index and can be used as a last resort

Austin, P. F., Bauer, S. B., Bower, W., et. al. (2016). The standardization of terminology of lower urinary tract function in children and adolescents: Update report from the standardization committee of the International Children's Continence Society. *Neurourology and urodynamics*, *35*(4), 471–481

DESMOPRESSIN

- **Mechanism:** antidiuretic hormone analogue → reduces urine formation
- **Indications:** enuresis in children (1st choice) and diabetes insipidus in patients ≥ 5 years (not for < 5 years). It is also indicated to treat nocturia in adults with ≤ 4 nocturnal voids (brand name Nocdurna®)
- It is available as oral tablets, melts (ODT), and nasal spray.
- **Contraindications:** hyponatremia (or predisposing conditions, e.g., cystic fibrosis, CKD, HF, diabetes, use of diuretics, or steroids), CrCl < 50ml/min, and acute illnesses associated with electrolyte imbalance (e.g., fever, diarrhea, or vomiting). Nasal spray is contraindicated in primary enuresis because of increased risk of hyponatremia (but can be used in diabetes insipidus)
- **Effect:** ↓ wet nights in 75% of patients **(target is > 50% drop in wet nights)**
- **Dose:** 1 tablet (120mcg) 1 hour before bedtime and ↑ every 3 days if no response (max. 360 mcg)
- **Onset:** it has a rapid onset – can be used PRN (e.g., for visits)
- **If effective, give drug holiday for 1-week q3m to assess the need for longer treatment.** There is spontaneous resolution in 15%/year
- **Adverse reactions:** Headache (most common), nausea, vomiting, ↑ BP (especially in elderly using nasal spray), tachycardia (caution in patients with angina), emotional disturbance, ↑ AST, and seizures. **Hyponatremia:** rare but serious. Limit the fluid intake to < 500 mL in children > 12 and < 250 ml in children < 12 within 1 hour before or 8 hours after the dose. Hold desmopressin in patients with acute illnesses causing electrolyte imbalance (e.g., fever, diarrhea, and vomiting). Patients should report early symptoms such as headache, nausea, and vomiting. Monitor the Na^+ level at baseline, 1 week and 1 month after starting therapy then periodically. Use with extreme caution in elderly
- **Drug interactions:**
    - SIADH causing drugs → ↑ effect of desmopressin
    - Demeclocycline and Li → ↓ effect of desmopressin

MALE SEXUAL DYSFUNCTION

Notes

Before initiating drug therapy, consider the patient's cardiovascular risk

	HTN	Ischemia	HF	Arrythmia	Valvular Dx	Other factors
Low Risk (Use drugs/ reassess)	Controlled	Stable/mild angina, successful PCA, MI > 6 weeks ago	NYHA I	Controlled A. Fib	Mitral prolapse	
Intermediate Risk (Evaluate before starting)	Controlled	Moderate/stable, MI 2 – 6 weeks ago	NYHA II			Stroke, PVD
High Risk (Defer therapy)	Uncontrolled	Unstable/refractory angina, MI < 2 weeks ago	NYHA III, IV	High risk	Moderate-severe	

Management of Delayed Ejaculation

- Sexual stimulation is essential and can be done by the partner (the patient should speak with the partner about the preferred method), or self-stimulation. The patient may use one or more types of stimulation (e.g., visual, physical, vibrator). Vaginal penetration should occur when the patient is close to ejaculate
- Review and manage medications that can cause delayed ejaculation including SSRI, antipsychotics, 5α reductase inhibitors, and alpha blockers

Management of Decreased Libido

- Male hypoactive sexual desire disorder (MHSDD) is defined as a persistent/recurrent lack of sexual thoughts, fantasies, and desire (DSM-5)
- Educate the patient and partner about response to sexual stimulation, motives, and barriers to stimulation (e.g., lack of stimulation, lack of communication about preferences, and feeling unloved)
- Evaluate and treat **comorbidities** such as depression (most important), hypogonadism, ED, anxiety, and sleep disorders, **biological factors** such as fatigue, age, shift work, and sedentary lifestyle, and **psychological factors** such as stress, anger, distraction, low self-esteem, self-monitoring, and fear of negative outcome
- Review and manage medications that can reduce sexual desire including **antidepressants** (SSRI, TCAD, MAOI), **AED** (phenytoin, CBZ, barbiturates), **drugs ↓ testosterone level/effect** (GnRH analogues, antiandrogen, spironolactone, 5-α reductase inhibitors, ketoconazole), beta blockers, opioids, and chronic alcoholic intake
- Testosterone can be useful in patients with MHSDD secondary to hypogonadism

FEMALE REPRODUCTIVE SYSTEM DISORDERS

Page 269

Management of Dysmenorrhea

- **Non-drug measures:** smoking cessation, regular exercise, yoga, application of heat, warm bath, and high frequency transcutaneous electrical nerve stimulation (TENS)
- **If seeking contraception:** CHC with EE ≤ 35mcg if not contraindicated (oral, transdermal, and vaginal CHC are the 1st line). Continuous and extended cycle are preferred
- If seeking contraception and estrogen is contraindicated OR dysmenorrhea is due to endometriosis: MPA
- **Dysmenorrhea + heavy menstruation:** levonorgestrel IUS, and etonogestrel implant
- **Not seeking contraception:** NSAID (start once pain starts and take regularly x 2-3 days). If NSAID is contraindicated → acetaminophen can be used but less effective. Ginger in doses of 750 mg – 2 g/day within 3 or 4 days of the beginning of the cycle has equal effect to NSAID
- **First choice:** CHC or NSAID x 3 – 6 months. If ineffective → use a combination x 3 – 6 months. If ineffective → reassess
- Pain persists + overlapping chronic pain (e.g., chronic pelvic pain) → add amitriptyline or gabapentin
- Surgical intervention for refractory cases
- Natural medicines: fenugreek, valerian, zinc, fish oil and/or vit. B1.

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Drug Measures

- **Vaginal estrogen** it restores vaginal pH, vaginal cells health, and vulvar/vaginal blood flow. It is useful when sexual disorders are due to menopause symptoms (e.g., vaginal dryness). Vaginal hyaluronic acid used 3x/week has similar benefit to estriol 0.5g cream
- **Vaginal lubricants** for dryness and pain due to dryness
- **Vaginal androstenolone (also known as dehydroepiandrosterone** or DHEA). It ↑ desire, ↑ coital comfort and ↑ sexual sensitivity, facilitates reaching orgasm, and improve dyspareunia. Effect is not due to increased estrogen level
- **Transdermal testosterone** can be used off-label for females with desire disorder, but studies did not show benefit and safety is not established
- **Sildenafil** improves the ability and time to reach orgasm in females with antidepressant-induced sexual dysfunction but does not improve arousal

NEW CONTENTS

GENITOPELVIC PAIN/PENETRATION DISORDERS

Diagnosis: Recurrence or persistence of ≥ 1 of the following symptoms for ≥ 6 months: 1- difficulty with vaginal penetration, 2- vulvovaginal/pelvic pain during penetration, 3- fear or anxiety of previous symptoms (1, and 2) in anticipation of, during, or because of vaginal penetration, and 4- tensing or tightening of pelvic flood muscles during an attempt of vaginal penetration

Definitions

- Dyspareunia: persistent/recurrent pain with attempted or complete vaginal penetration/entry
- Vaginismus: persistent/recurrent difficulty to allow vaginal entry of penis, finger, or object
- Provoked vestibulodynia (PVD): pain from touching vaginal opening associated with allodynia
- Genitourinary Syndrome of Menopause (GSM): dyspareunia associated with menopause symptoms (labia atrophy, loss of elasticity)

Drug Measures

- Vaginal estrogen, vaginal hyaluronic acid, vaginal lubricants/moisturizers, and vaginal endrostenolone (DHEA)
- Topical testosterone and topical estrogen applied to posterior fourchette for recurrent tearing
- Topical sodium cromoglycate to prevent the release of inflammatory mediators
- Introital lidocaine for patients who continue to experience allodynia despite psychotherapy → anticipation of less pain and ↓ intensity of dyspareunia

MINOR EYE DISORDERS

Page 274

Acute bacterial conjunctivitis
Symptoms: red, irritated eye + purulent discharge
Non-Drug Measures: handwashing, avoid eye-to-eye contact, warm compress to remove morning crust, and eye irrigation. Refer children and patients wearing contact lenses to the physician. Keep child out of school for 24 hours after staring therapy. It is very contagious.
Drug Therapy
- Most cases are self-limiting/benign. Antibiotics reduce the recovery time. Antibiotic therapy can be delayed x 3 days to ↓ antibiotic use and resistance (the delay prolongs the duration by only ½ day)
- **Mild cases:** Polysporin® (polymyxin B sulfate + gramicidin) eye drops, 1-2 drops into the affected eye QID x 7 days (continue x 2 days after symptoms resolution). If no improvement in 3-5 days, regardless of whether treated or not, refer to the physician
- **Severe cases:** prescription antibiotics e.g., trimethoprim + polymyxin B (Polytrim®) or tobramycin. FQ (besi, cipro, gati, moxi, and o-floxacin) are reserved for severe cases

Page 275

Management of viral conjunctivitis
- Patient experiences foreign body sensation with watery + non-itchy eye (suggest viral eye infection).
- Causative agents:
  - **Adenovirus (the most common):** usually bilateral or one eye is infected few days after the other. Transmitted from respiratory secretions to the eye by touch or from contaminated objects or swimming pools. Incubation is 2 – 14 days and the duration is 2 – 4 weeks. Case remains contagious for 2 weeks after the 2nd eye involvement
  - **Herpes simplex virus (HSV):** unilateral water discharge + eyelid vesicles
  - **Herpes zoster:** involved conjunctiva and eyelid + uveitis, corneal complications, and facial vesicles
- **Non-Drug Measures:** cold compress, keep patients at home for 1 week (until no eye discharge) and avoid direct contact with others for 14 days
- **Drug Therapy:** symptomatic treatments: **lubricants**, **antihistamines** (ketotifen, olopatadine), and **decongestants** (naphazoline, oxymetazoline, phenylephrine, or tetrahydrozoline – used for ≤ 3 days or ≤ 3-4 times/month to avoid hyperemia. Avoid in patients with angle-closure glaucoma). For **adenovirus**, antivirals are NOT recommended, and steroids can prolong viral shedding but can be used post-infection for visual opacity. For **HSV**, topical trifluridine (1 drop q2h while awake for a maximum of 9x/day until re-epithelization then q4h for a maximum of 5x/day for 7 days). Oral antivirals can be used for HSV infections
- **Note:** patients with non-HSV conjunctivitis commonly have sore throat and lymphadenopathy and no discharge. Antibiotics can be used to prevent secondary infections

NEW CONTENTS

Notes

- A combination of antibiotic + corticosteroid should be reserved for severe blepharitis, rosacea-associated eyelid disease, post-surgical or severe conjunctivitis. Corticosteroids may worsen infectious conjunctivitis but can be used for ≤ 1 week in severe cases.

GLAUCOMA

See the minor updates in the update video

CATARACT

Page 279

Steroids

- Used to ↓ inflammation and macular edema
- **Agents:** loteprednol (Lotemax®), difluprednate (Durezol®), rimexolone (Vexol®), prednisolone, fluorometholone, and dexamethasone.
- **Use:** usually start **POST-OP** and continue for 3 to 4 weeks (difluprednate starts 24-hour pre-op, post-op x 2 weeks, then taper). Longer duration is required if the patient develops cystoid macular edema

NSAID

- Used to ↓ inflammation and macular edema
- **Agents:** bromfenac (Prolensa®), nepafenac (Nevanac® and Ilevro®), and ketorolac (Acular®)
- **Use:** used **PERI-OP** (start **1 day before** and x 2 weeks **post-op**. Diclofenac and ketorolac are used for 3-4 weeks). Longer duration is required if the patient develops cystoid macular edema

ARMD

Page 281

ADD – a new VEGF inhibitor – brolucizumab

MINOR EYE DISORDERS

No Updates

SYSTEMIC LUPUS

No Updates

HEAT RELATED DISORDERS

No Updates

ADVERSE REACTIONS OF CHEMOTHERAPIES

Please see the new chapter and video on the portal under "Cancer Care"

ANEMIA

No Updates